

SARAPUL, Russian Federation

WMO: 284180

Lat: 56.467N

Lon: 53.733E

Elev: 437

StdP: 14.47

Time Zone: 4.00 (E04)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.4	-12.4	-22.4	1.6	-17.0	-17.2	2.2	-12.1	17.0	16.4	14.5	15.6	2.5	0	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	88.0	70.0	84.5	68.1	81.1	66.6	71.5	85.0	69.7	81.6	67.9	78.7	5.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	100.1	77.4	65.2	94.6	76.1	63.5	89.2	74.6	35.5	85.6	33.9	81.6	32.4	78.6	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.0	11.6	DB	-24.6	91.1	6.3	3.6	-29.1	93.7	-32.8	95.9	-36.3	97.9	-40.9	100.6	(4)
(5)			WB	-24.7	73.9	6.3	1.9	-29.3	75.2	-32.9	76.3	-36.5	77.4	-41.0	78.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	8.4	12.5	23.8	40.7	56.6	63.6	67.0	64.3	53.3	39.0	26.9	15.7		
	DBStd	22.67	12.10	11.53	8.72	8.72	9.12	7.95	6.80	8.63	7.26	8.21	9.59	12.27		
	HDD50	5660	1290	1051	812	305	40	3	0	2	46	354	693	1064		
	HDD65	9652	1755	1471	1277	729	289	119	58	120	356	806	1143	1529		
	CDD50	1814	0	0	0	26	244	412	528	447	144	13	0	0		
	CDD65	330	0	0	0	0	28	78	120	99	5	0	0	0		
	CDH74	3065	0	0	0	4	298	680	1107	939	37	0	0	0		
	CDH80	953	0	0	0	0	62	177	354	357	3	0	0	0		
(14) Wind	WSAvg	5.1	5.1	4.8	5.4	5.5	5.4	5.1	4.7	4.6	4.7	5.3	5.1	5.0		
(15) Precipitation	PrecAvg	21.6	1.5	1.1	1.0	1.2	1.7	2.4	2.3	2.5	2.1	2.1	1.8	1.7		
	PrecMax	28.8	3.0	3.4	3.5	3.5	4.8	4.8	5.9	7.2	5.2	4.1	4.2	4.6		
	PrecMin	12.7	0.3	0.0	0.0	0.2	0.1	0.5	0.0	0.3	0.0	0.2	0.6	0.1		
	PrecStd	3.6	0.6	0.6	0.7	0.7	1.1	1.3	1.2	1.4	1.3	0.9	0.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	35.0	44.1	73.5	85.7	88.6	92.9	93.8	79.0	63.3	47.9	36.0		
		MCWB	31.7	32.9	37.5	58.6	65.7	70.4	73.0	72.8	62.1	52.5	46.0	34.8		
	2%	DB	31.6	33.3	39.6	66.6	81.3	84.9	88.0	88.3	72.7	57.4	45.1	34.1		
		MCWB	30.3	31.7	34.7	54.3	63.9	69.5	70.8	69.2	60.0	49.9	43.4	33.0		
	5%	DB	28.9	31.2	37.1	59.8	77.1	81.4	83.9	83.7	68.5	53.8	42.2	32.7		
		MCWB	27.2	29.6	33.5	48.9	61.6	67.3	68.0	68.2	57.7	48.3	40.8	31.6		
	10%	DB	25.0	27.3	35.1	54.8	72.9	77.7	80.3	78.9	65.0	50.4	37.8	30.3		
		MCWB	23.7	25.7	32.5	46.8	59.1	65.1	66.8	66.1	56.3	46.7	36.5	28.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.7	33.7	39.1	59.6	68.0	72.6	74.1	73.4	63.9	54.9	46.5	35.1		
		MCDB	33.3	34.4	42.6	72.4	80.6	84.9	89.2	90.8	72.6	60.3	47.6	35.4		
	2%	WB	30.3	32.1	35.8	54.7	65.2	70.7	71.7	70.8	61.5	51.2	44.0	33.3		
		MCDB	31.7	33.1	38.4	64.1	78.8	82.7	85.3	84.7	69.8	55.5	45.2	33.9		
	5%	WB	27.5	29.3	34.3	50.4	62.8	68.7	69.8	68.9	59.4	49.0	40.6	31.6		
		MCDB	28.9	30.9	36.2	58.8	75.6	79.6	80.7	81.4	66.2	52.8	42.1	32.6		
	10%	WB	23.7	25.8	32.8	46.7	59.9	66.2	67.8	66.8	57.4	47.0	36.6	29.2		
		MCDB	24.9	27.2	34.8	54.4	71.1	75.5	77.8	77.5	63.3	50.5	37.6	30.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	11.2	12.7	14.4	19.7	17.7	17.7	16.9	14.3	9.5	6.9	8.9		
		MCDBR	8.7	7.3	11.5	23.0	24.5	21.7	23.3	22.8	21.1	15.3	7.3	6.7		
	5% WB	MCWBR	8.6	6.7	8.2	13.2	12.7	10.3	10.6	9.8	11.1	9.7	7.1	6.6		
		MCWBR	8.5	6.8	10.2	21.4	22.8	19.6	21.4	20.6	18.5	12.8	7.3	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.318	0.373	0.405	0.392	0.388	0.409	0.429	0.376	0.353	0.313	0.290		
	taud		2.022	2.025	1.945	2.094	2.292	2.356	2.305	2.223	2.344	2.271	2.139	1.999		
	Ebn at Noon		174	224	239	254	268	271	262	246	244	213	169	140		
	Edh at Noon		20	31	43	43	38	36	37	38	29	24	18	16		
(44) All-Sky Solar Radiation	RadAvg		211	515	970	1379	1708	1824	1810	1380	857	381	164	117		
	RadStd		29	68	93	141	164	156	174	126	97	72	33	28		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (12 neighbors)	+1.03	N/A	N/A	N/A	N/A	N/A	N/A	-362	N/A	N/A	(47)

Nomenclature: See separate page